



ANALYTICAL ABILITY & LOGICAL REASONING

- A set consecutive positive integers beginning with 1 is written on the blackboard. A student came along and erased one number. The average of the remaining numbers is 35717. What was the number erased?
(a)7 (b)8 (c)9 (d)None
- Four friends A, B, C and D need to cross a bridge in the night. A maximum of 2 people can cross at a time . they have only one lamp. A takes one min to cross the bridge. B Take 2 min , C takes 8 min and D takes 11 min to cross the bridge respectively. A pair must walk together at the speed of the person who walks slowly. What is the minimum time required to cross the bridge by all the four people?
(a)23 min (b) 20 min (c) 18 min (d) 16 min
- In a city, 40.1% of the adults are illiterate while 85.1% the children are literate. If the ratio of the adults to that the children is 2:3, then what percent of the population is literate?
(a)20% (b)25% (c)50% (d)75%
- A runs $1\frac{2}{3}$ times as fast as B. If A gives B a start of 80m, how far must winning post be so that A and B might reach it at the same time?
(a)200m (b)400m (c)300m (d)160m
- A person's present age is two fifth of the age of his mother. After 8 year, he will be one half of the (age) of his mother. What is the present age of his mother?
(a) 60yr (b)50yr (c)40yr (d)30yr
- Mr. Kumar drives to work at an average speed of 48 km/hr. the time taken to cover the first 60% of the distance is 10 min more than the time taken to cover the remaining distance. How far is his office ?
(a)30km (b)40km (c)45km (d)48km
- Two pipes A and B can fill the cistern in 37.5 min and 45 min respectively. Both pipes are opened. The cistern will be filled in just half an hour, if the B is turned off after
(a) 5 min (b) 9 min (c)10 min (d)15 min
- In a certain code, DOES is written as 5\$3% and SITE is written as %4#3. How is EDIT written in that code?
(a)3#4\$ (b)%3#5 (c)354# (d)4#5\$
- In a shower, 5cm of rain falls. The volume of water that falls on 1.5 hectares of ground is
(a)75m³ (b)750m³ (c)7500m³ (d)75000m³
- Eight friends A through H, are sitting around a circular table, playing a game of cards. They belong to two different teams X and Y. No two persons of the same team sit in adjacent seats.
 - A sits neither opposite to D nor to H but is sitting in between C and G.
 - B sits neither opposite to A nor to G but is sitting in between F and D.
 - B and H belong to team X and D sits opposite to E
 Who are the members of team X?
(a) A, D, F and E (b) B, H, C and E
(c) B, D, H and G (d) B, H, C and G
- Eight friends A through H, are sitting around a circular table, playing a game of cards. They belong to two different teams X and Y. No two persons of the same team sit in adjacent seats.
 - A sits neither opposite to D nor to H but is sitting in between C and G.

- B sits neither opposite to A nor to G but is sitting in between F and D
- B and H belong to team X and D sits opposite to E Who are sitting adjacent to E?
(a)B and H (b)B and G (c)H and G (d)H and C
- Four students A, B, C and D distributed 30 marbles among themselves. No two students got equal number of marbles. No students got more than 10 marbles. No student got less than 5 marbles. A and C got odd number of marbles. B and D got even number of marbles. A got more marbles than B,C got more marbles than D,B got more marbles than D.
What is the number of marbles with A?
(a) 6 (b)7 (c)8 (d)9
- Four students A, B, C and D distributed 30 marbles among themselves. No two students got equal number of marbles. No students got more than 10 marbles. No student got less than 5 marbles. A and C got odd number of marbles. B and D got even number of marbles. A got more marbles than B,C got more marbles than D,B got more marbles than D.
Mean of number of marbles with B, C, D is
(a)6 (b)7 (c)8 (d) None
- Nine individuals-Z, Y, X, W, V, U, T, S and R- are the only candidates, who can serve on three committees-k₁, k₂ and k₃, and each candidate should serve on exactly one of the committees. Committee k₁ should consist of exactly one number more than committee-k₂. It is possible that there are no members in committee k₃. Among Z, Y and X none can serve on committee k₁. Among W,V and U none can serve on committee k₂. Among T, S and R none can serve on committee k₃.
In case committee K₂ is served by T and Z only, how many of the nine individuals should serve on committee k₃?
(a)3 (b)4 (c)5 (d)6
- Nine individuals – Z, Y, X, W,V,U,T,S and R- are the only candidates, who can serve on three committees- K₁,K₂ and K₃, and each candidate should serve on exactly one of the committees. Committee K₁ should consist of exactly one member more than committee K₂. It is possible that there are no members in committee K₃. Among Z, Y and X none can serve on committee K₁, Among W, V and U none can serve on committee K₂.
Among T, S and R none can serve on committee K₃.
Of the nine individuals, the largest number that can serve together on committee K₃ is
(a)8 (b)7 (c)6 (d)5
- Fill in the blank in the series ELFA, GLHA, ILJA,_____. MLNA:
(a)OLPA (b)KLMA (c)LLMA (d)KLLA
- Pointing to a gentleman, Mohan said, 'His only brother is the father of my daughter's father'. The gentleman is Mohan's
(a)brother (b)father (c)uncle (d) None
- It was 9.35 am in Garvita 's watch, which kept correct time, when Manya informed her that the last bus left the bus stop at 9.25 am. Manya' s watch is 5 min fast. The frequency of the bus is every 20 min. For how long Garvita must wait to catch the next bus?
(a)5 min (b)10 min (c)15 min (d)20 min

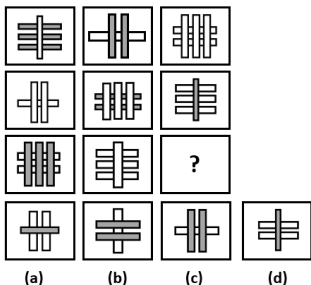




19. A total of 324 notes comprising of ₹20 and ₹ 50 denomination make a sum of ₹ 12450. The number of ₹ 20 notes is
(a)200 (b)144 (c)125 (d)110

20. Rishabh stops after going 10 km towards west from his office. Then he goes 8 km turning to his left. After this he goes 4 km turning to his left. How far is he from the fixed point?
(a)18 km (b)8 km (c)10 km (d)None

21. Which of the four options should fill the missing cell?



- (a)1 (b)2 (c)3 (d)4

22. In the following questions, the symbols \$, #, @, % and * illustrate the following meanings.

- >P\$Q – P is not smaller than Q
> P#Q –P is neither greater than nor equal to Q.
>P@Q –P is neither smaller than nor equal to Q.
>P%Q- P is not greater than Q
>P*Q –P is neither greater than nor smaller than Q.

Statements: K#L, L%M, M*N, N#O

Conclusions:

- I. K#M II. K*M III. L%O
(a)I only (b)Either I or II only
(c)III only (d)All I, II and III

23. Study the following arrangement carefully and answer the question given below:

W1R % 4JE # 7MT2I9BH3A\$9FQ5DG6USP

Three of the following are alike in a certain way on the basis of above arrangement and hence form a group.

Which one does not belong to that group?

- (a)R W 4 (b)9 Q A (c)3 B \$ (d)5 F G

24. If there are no dancers that aren't slim no singers that aren't dancers, then which statements are always true?

Choose the correct answer.

- (a) There is not one slim person that is not a dancer.
(b) All singers are slim.
(c) Anybody slim is also a singer.
(d) None of the above

25. If in a certain language, ITNIETAM is the code for INTIMATE, which word has the code TREVNIETARBI?

- (a)INVERTIBRATE (b)INVERTIBARTE
(c)INVERTIBRATE (d)INVERTIBRETA

26. Sum of ages of Anu and bhanu is 10 yr more than sum of ages of Bhanu, Chanu and Dhanu. Average age of Chanu and dhanu is 19 yrs . Find the average age of Anu and Dhanu if Dhanu is 10 years elder than Chanu.

- (a)36 yr (b)30 yr (c)25 yr (d)31 yr

27. In a competitive examination in Maharashtra state 9% candidates got selected from the total appeared candidates. Tripura state had an equal number candidates appeared and 12% candidates got selected with 102 more candidates got

selected than Maharashtra state. What was the number of candidates appeared from each state?

- (a)3400 (b)3000 (c)2850 (d)3200

28. Shiva gave 40% of his monthly salary to his mother from the remaining he used 7% for electronic gadgets and 23% he kept aside for his monthly expenses. The remaining amount he transferred to his friend's account. The sum of the amount he gave to his mother and he transferred to his friend account was ₹ 41000. What was Shiva's monthly salary?

- (a)₹ 50500 (b)₹ 49000 (c)₹50000 (d)₹ 45000

29. Read the information given below and answer the questions that follow:

- (i) A*B means→ A and B are of the same age
(ii) A-B means → B is younger than A
(iii) A+ B means→ A is younger than B
Sachin *Madan- Reena means?

- (a) Reena is youngest
(b) Reena is oldest
(c) Madan is younger than Reena
(d) Madan is youngest

30. Read the information given below and answer the questions that follows

- (i) A*B means→ A and B are of the same age
(ii) A-B means → B is younger than A
(iii) A+ B means→ A is younger than B
X +Y+Z is same as.....?

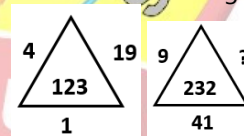
- (a)Y-X-Z (b)Z-Y-X (c)Z-X-Y (d)X-Y-Z

31. Find out the wrong number in the following number series:

56, 58, 62, 70, 84, 118, 182

- (a)58 (b)62 (c)84 (d)118

32. Find out the missing number



- (a)2 (b)6 (c)4 (d)16

33. In a examination, 78% of the total students who appeared were successful. If the total number of failures was 176 and 34% got first class, then how many students got first class?

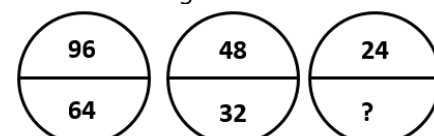
- (a)272 (b)112 (c)210 (d)254

34. Which number should come in place of the question mark(?) in the following chart:

1	7	9
2	14	?
3	105	117

- (a)16 (b)26 (c)20 (d)12

35. Find the missing number



- (a)21 (b)16 (c)10 (d)8

36. How many minimum numbers of colours will be required to paint all the sides of a cube without the adjacent sides having the same colours?





MAARULA CLASSES

TARGET- NIMCET / CUET.PG By: Amit Katiyar (MCA-JNU)

NIMCET ORIGINAL
PAPER 2020
PYQ No: 02



SCAN QR OF OUR APP MAARULA MATHEM

- (a)3 (b)4 (c)5 (d)6
37. If a man walks at the rate of 4km/hr, he misses a train by only 6 min. However, if he walks at the rate of 5km/hr, he reaches the station 6 min before the arrival of the train. The distance covered by him to reach the station is
- (a)4 km (b)7 km (c)9 km (d)5 km
38. If the numerator of a fraction is increased by 25% and denominator decreased by 20%, the new value is $\frac{5}{4}$. What is the original value?
- (a) $\frac{3}{5}$ (b) $\frac{4}{5}$ (c) $\frac{7}{8}$ (d) $\frac{3}{7}$
39. Read the following information carefully and answer the questions given below
- (i) Five friends Amar, Kapil, Sarvesh, Rohan and Nagesh put on five shirts of different colour, i.e., Red, Yellow, Blue, White, and Green, while they were going to attend to party. These colours are not in the order.
- (ii) They have different hobbies as reading, playing, outing, singing and writing.
- (iii) Kapil, who likes singing, does not wear Yellow shirt. Sarvesh wears Red shirt and he does not like reading or writing. Nagesh likes playing and he does not wear Blue or Yellow shirt. Amar likes writing and Rohan does not wear Yellow or Green shirt.

What is the colour of Kapil's shirt?

- (a)White (b)Green
(c)Blue (d)Insufficient data to answer

40. Read the following information carefully and answer the questions given below

- (i) Five friends Amar, Kapil, Sarvesh, Rohan and Nagesh put on five shirts of different colour, i.e., Red, Yellow, Blue, White, and Green, while they were going to attend a party. These colours are not in a order.
- (ii) They have different hobbies as reading, playing, outing, singing and writing.
- (iii) Kapil, who likes singing, does not wear Yellow shirt. Sarvesh wears Red shirt and he does not like reading or writing. Nagesh likes playing and he does not wear Blue or Yellow shirt. Amar likes writing and Rohan does not wear Yellow or green shirt.

Who likes writing?

- (a) Rohan (b)Amar
(c) Kapil (d) Insufficient data to answer

ANSWER

1.	A	2.	C	3.	D	4.	A	5.	C
6.	B	7.	B	8.	C	9.	B	10.	D
11.	D	12.	D	13.	B	14.	B	15.	C
16.	D	17.	C	18.	A	19.	C	20.	C
21.	A	22.	A	23.	B	24.	D	25.	C
26.	A	27.	A	28.	C	29.	A	30.	B
31.	C	32.	C	33.	A	34.	D	35.	B
36.	A	37.	A	38.	B	39.	D	40.	b





SOLUTION

- (a) Since all the values are integers, their sum must be an integer.
Sum = Number $\times$ Average
Number of integers after erasing must be a multiple of 17.
As the average of n consecutive numbers is $\left(\frac{n+1}{2}\right)$, then number of integers must be around 69.
After erasing the number of integers is 68 (as 68 is a multiple of 17) and before erasing, the number of integers is 69.
Let the erased number be x .
$$\frac{69 \times 35 - x}{68} = \frac{602}{17}$$
$$X = 7$$
- (c)
First A and B will go in 2 min, then A returns in 1 min.
Then C and D will go in 11 min. B returns in 2 min.
A & B will go in 2 min.
The total time taken would be
(2 + 1 + 11 + 2 + 2) = 18 min.
- (d) Let the number of adults is $2x$ and number of children be $3x$.
Number of literate adults = 60% of $2x = \frac{6x}{5}$
Number of literate children = 85% of $3x = \frac{51x}{20}$
Total literate population = $\frac{6x}{5} + \frac{51x}{20} = \frac{75x}{20}$
% of literate = $\frac{75x}{20} \times \frac{100}{5x}$
 $\Rightarrow 75\%$
- (a) We have,
Speed of A : Speed of B = $\frac{5}{3} : 1 = 5 : 3$
 $\Rightarrow$ In a race of 5 meters, A runs 5 meters and B runs 3 meters.
 $\Rightarrow$ A gains 80 metres over B in a race of $\frac{5}{2} \times 80 = 200$ m
- (c) Let the mother's present age be x years.
Then, the person's present age = $\frac{2}{5}x$ years
According to questions,
$$\left(\frac{2x}{5} + 8\right) = \frac{1}{2}(x + 8)$$
$$\Rightarrow \frac{2x+40}{5} = \frac{1}{2}(x + 8)$$
$$\Rightarrow 4x + 80 = 5x + 40$$
$$\Rightarrow x = 40$$
 yr
- (b) Let the total distance be x km.
Time taken to cover remaining 40% of x distance is
$$t_1 = \frac{40 \times x}{100 \times 48}$$

Since, time taken to cover first 60% of x distance is $t_2 = t_1 + \frac{10}{60}$ hrs.
$$\Rightarrow t_2 = \frac{60 \times x}{100 \times 48}$$
$$\therefore \frac{60 \times x}{100 \times 48} = \frac{40 \times x}{100 \times 48} + \frac{10}{60}$$
$$\Rightarrow \frac{100 \times 48}{60 \times x} = \frac{100 \times 48}{40 \times x} + \frac{60}{10}$$
$$\Rightarrow \frac{100 \times 48}{20x} = \frac{100 \times 48}{10x} + \frac{60}{10}$$
$$\Rightarrow \frac{4800}{20} = \frac{4800}{10} + \frac{60}{10}$$
$$\Rightarrow x = \frac{10}{60} \times \frac{4800}{20}$$
$$\Rightarrow x = 40$$
 km.
- (b) Let B turned off after x minutes. Then, part filled by (A+B) in x min. + part filled by A is (30-x) min. = 1

$$\therefore x \left(\frac{2}{75} + \frac{1}{45} \right) + (30 - x) \frac{2}{75} = 1$$

$$\Rightarrow \frac{11x}{225} + \frac{60-2x}{75} = 1$$

$$\Rightarrow 11x + 180 - 6x = 225$$

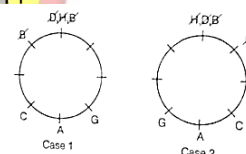
$$\Rightarrow 5x + 180 = 225$$

$$\Rightarrow x = 9 \text{ min.}$$

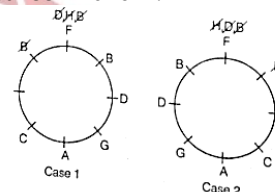
- (c) As per the given questions,
DOES
5\$3%
SITE
%4#3
Edit is coded as 354#
 $\therefore$ The correct answer is option (c).

- (b) 1 hectare = 10000 m²
So, Area = (15 $\times$ 10000) m²
= 15000 m²
Depth = $\frac{5}{100} \text{ m} = \frac{1}{20} \text{ m}$
Volume = (Area $\times$ Depth)
= $\left(15000 \times \frac{1}{20}\right) \text{ m}^3$
= 750 m³
= 750 cubic metre.
 $\therefore$ The correct answer is option (b).

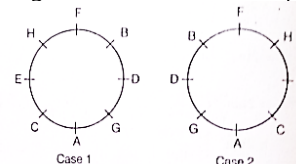
- (d) Given person : A, B, C, D, E, F, G and H
(i) A sits neither opposite of D nor to H but is sitting in between C and G.
(ii) B sits neither opposite to A nor to G.



B is sitting in between F and D.

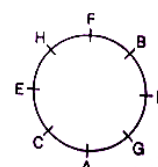


- (iii) B and H belong to team X and D sits opposite to E.



As B and H belong to team X, so is C and G.
Remaining F, E, A and D belong to team Y.
Hence, B, H, C and G is the correct answer.

- (d) As per the question, the arrangement is



- $\therefore$ H and C are sitting adjacent to E.
 $\therefore$ The correct answer is option (d).





12. (d) Given 4 students A, B, C, D
 $\Rightarrow$ They distributed 30 marbles among themselves.
 (a) No two students got equal number of marbles.
 (b) No student got more than 10 marbles.
 (c) No student got less than 5 marbles.
 (d) A and C got odd number of marbles. It can be 7 or 9.
 (e) B and D got even number of marbles. It can be 6, 8 or 10.
 (f) A got more marbles than B, C got more marbles than D
 $A > B > D; C > D$

$$A(7,9) > B(6,8) > D(6,8), C(7,9) > D(6,8)$$

So, D got the least number of marbles.

$\therefore$ D got 6 marbles.

So, B got 8 marbles as B got more marbles than D.

$\Rightarrow$ A got 9 marbles as A got more marbles than B.

$\Rightarrow$ C got 7 marbles as C got more marbles than D.

Students	Definite Number of marbles
A	9
B	8
C	7
D	6

$\therefore$ The number of marbles with A is 9.

$\therefore$ The correct answer is option (d).

13. (b) As per the question, the different number of marbles.

Students	Definite Number of marbles
A	9
B	8
C	7
D	6

Mean of number of marbles with B, C, D is

$$\frac{8+7+6}{3} = \frac{21}{3} = 7$$

$\therefore$ The correct answer is option (b).

14. (b) Given, Nine individuals : Z, Y, X, W, V, U, T, S and R.

Three committees : K_1 , K_2 and K_3

1. Each candidate should serve on exactly one of the committees.

2. Among Z, Y and X none can serve on committee K_1 .

3. Among T, S and R none can serve on committee K_3 .

4. Among W, V and U none can serve on committee K_2 .

In case committee K_2 is served by T and Z.

Committees	Individual who can serve	Individuals who can't serve
K_1		Z,Y,X
K_2	T,Z	W,V,U
K_3		T,S,R

K_1 should consist of exactly one member more than committee K_2 .

So, K_1 is served by three members as K_2 is served by two members only.

So, remaining individuals will serve committee K_3 .

$$\text{Remaining persons} = 9 - (2 + 3)$$

$$= 9 - 5 = 4$$

The correct answer is option (b).

15. (c) As per the given question,

Committees	Individual who can serve	Individuals who can't serve
K_1		Z,Y,X
K_2		W,V,U
K_3		T,S,R

It is possible that there are no members in committee K_3 .

Of the nine individuals, the largest number that can serve together on committee $K_2 = 9 - \text{Number of members who can't serve } K_3$.

$$= 9 - 3 = 6$$

$\therefore$ The correct answer is option (c).

16. (d) According to the given question.

First letter of each term

$$E \xrightarrow{+2} G \xrightarrow{+2} I \xrightarrow{+2} K \xrightarrow{+2} M$$

Second letter of each term, is same

$$L \rightarrow L \rightarrow L \rightarrow L \rightarrow L$$

Third letter of each term:

$$F \xrightarrow{+2} H \xrightarrow{+2} J \xrightarrow{+2} L \xrightarrow{+2} N$$

Fourth letter of each term (is same)

$$A \rightarrow A \rightarrow A \rightarrow A \rightarrow A$$

$\therefore$ The series in the blanks is KLLA.

$\therefore$ The correct answer is option (d).

17. (c) In the diagram, '+' shows male.

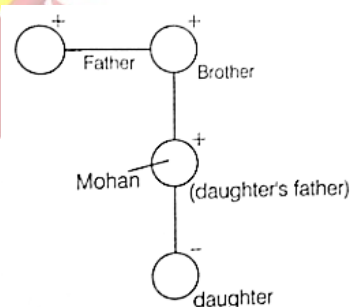
'-' shows female.

Vertical line shows the son / Father.

Horizontal line shows the brother / sister.

Double horizontal line shows the couple.

According to the question.



The gentleman is uncle to Mohan.

$\therefore$ The correct answer is option (c).

18. (a) It was 9.35 AM in Garvita's watch, which kept correct time.

$\Rightarrow$ Manya informed her that the last bus left the bus stop at 9.25 am. Manya's watch is 5 min fast.

$\Rightarrow$ So, the actual time when the bus left the bus stop.

$$= 9.25 - 0 : 05 \text{ AM} = 9:20 \text{ AM.}$$

$\Rightarrow$ The frequency of the bus is every 20 min.

So, the next bus will arrive at bus stop at 9:40 AM.

$\Rightarrow$ As it was 9.35 AM in Garvita's watch, she should wait, 5 more minutes to catch the next bus.

$\therefore$ The correct answer is options (a).

19. (c) Given, 324 notes comprised of Rs. 20 and Rs. 50 total sum is Rs. 12450.





Let the number of Rs. 20 notes = x

The number of Rs. 50 notes = y

According to the question,

$$x + y = 324 \quad \dots\dots\dots (i)$$

$$20x + 50y = 12450 \quad \dots\dots\dots (ii)$$

Solving Eqs. (i) and (ii) we get,

$$1 \times \text{Eq. (i)} \quad 20x + 50y = 12450$$

$$20x + 20y = 6480$$

$$20x \quad \text{Eq. (ii)}$$

$$30y = 5970$$

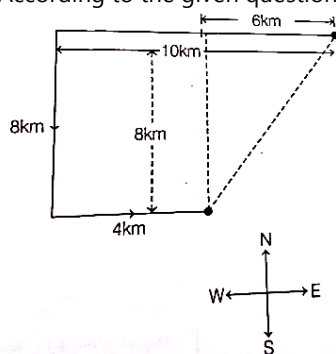
$$y = 199$$

$$\therefore x = 125$$

$\therefore$ Number of Rs. 20 notes is 125

$\therefore$ The correct answer is option (c).

20. (c) According to the given questions.



Distance between his starting point and final point

$$\begin{aligned} \text{Apply Pythagoras theorem} &= \sqrt{8^2 + 6^2} \text{ km} \\ &= \sqrt{64 + 36} \\ &= \sqrt{100} = 10 \text{ km.} \end{aligned}$$

$\therefore$ The correct answer is option (c).

21. (a) We can see from the given figure.
From 1st row, the format of horizontal lines are 3,1,2 and the format of vertical lines are 1,2,3
From 2nd row, the format of horizontal lines are 1,2,3 and the format of vertical lines are 2,3,1
From 3rd row, the format of horizontal lines must be 2,3,1 and the format of the vertical line must be 3,1,2
In the third column, only strip must be shaded, so Option (a) is correct answer.

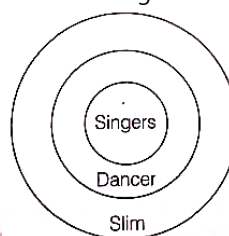
22. (a) As per the given data. $K \# L$ means K is neither greater than nor equal to L. $L \% M$ means L is not greater than M, it means L is smaller or equal to M. $M * N$ means M is neither greater than nor smaller than N, it means M is equal to N. $N \# O$ means N is neither greater than nor equal to O, it means N is smaller than O.
 $K < L \leq M = < O$
By this, $K \# M$ is correct.
 $K * M$ is incorrect.
 $L \% O$ is incorrect.
 $\therefore K \# M$ means 1 only correct.
 $\therefore$ The correct answer is option (a).

23. (b) In all except, 9Q A the second element is second to the left of first element and third element is second to the right of first element.

W 1 R % 4 J E # 7 M T 2 1 9 B H 3 (A) \$ (9) F (Q) 5 D G 6 U S P

$\therefore$ The correct answer is option (b).

24. (d) According to the given statements,
If there are n dancers that aren't slim
All dancers are slim
No singers that aren't dancers
All singers are dancers.
The Venn diagram of above two statements are



Statement I $\rightarrow$ False

Statement II $\rightarrow$ True

Statement III $\rightarrow$ False

Statement IV $\rightarrow$ False

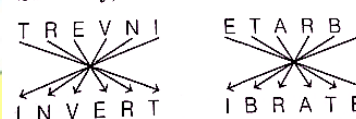
$\therefore$ The correct statement is "All singers are slim".

$\therefore$ The correct answer is option (d).

25. (c) As per the given questions.



Similarly,



Similarly,

$\therefore$ The correct answer is option (c).

26. (a) Sum of ages of Anu and Bhanu is 10 years more than sum of ages of Bhanu, Chanu and Dhanu,
 $\text{Anu} + \text{Bhanu} = (\text{Bhanu} + \text{Chanu} + \text{Dhanu}) + 10$
 $\text{Chanu} + \text{Dhanu} + 10 = \text{Anu} \quad \dots\dots\dots (i)$
Average age of Chanu and Dhanu is 19 years
 $\frac{(\text{Chanu} + \text{Dhanu})}{2} = 19$
 $\text{Chanu} + \text{Dhanu} = 38 \text{ year} \quad \dots\dots\dots (ii)$
Putting the value of (ii) in (i) we get,
 $38 + 10 = \text{Anu}$
So, age of Anu is 48 years
Dhanu is 10 years elder than Chanu
 $\text{Dhanu} - 10 = \text{Chanu}$
Putting the value of (iii) in (ii), we get
 $\text{Dhanu} - 10 + \text{Dhanu} = 38$
 $2 \text{ Dhanu} = 38 + 10$
 $2 \text{ Dhanu} = 48$
 $\text{Dhanu} = 24 \text{ yrs.}$
Average age of Anu and Dhanu $(48 + 24)$
 $= \frac{72}{2} = 36 \text{ years}$
 $\therefore$ The answer is 36 yr.
 $\therefore$ The correct answer is option (a).



27. (a) Given,

⇒ Selected candidate in Maharashtra from the total appeared candidates = 9%

⇒ Selected candidate in Tripura from the total appeared candidates = 12%

⇒ In Tripura number of selected candidate more than that of Maharashtra = 102

Let total number of candidates from each state be 100 unit.

Selected candidate in Maharashtra from the total appeared candidates = $\left(\frac{9}{100}\right) \times 100 = 9 \text{ Unit}$

⇒ Selected candidate in Tripura from the total appeared candidates = $\left(\frac{12}{100}\right) \times 100 = 12 \text{ Unit}$

In Tripura number of selected candidates more than that of Maharashtra = 102

⇒ (12-9) unit = 102

⇒ 3 unit = 102

⇒ 1 unit = 102/3

⇒ 1 unit = 34

(1 x 100) unit = 34 x 100

⇒ 100 unit = 3400

∴ The number of candidates appeared from each state was 3400.

∴ The correct answer is option (a).

28. (c) Salary given to his mother = 40% = $\frac{2}{5}$ from the remaining, he used for electronic gadgets and monthly expenses = $(7 + 23)\% = 30\% = \frac{3}{10}$

Sum of the amount he gave to his mother and transferred to his friend accounts = Rs. 41000

Let the sum of amount be 50 unit (where 50 = product of denominator of given fraction 5×10)

Salary given to his mother = $\left(\frac{2}{5} \times 50\right)$

= 20 unit

Remaining salary after giving to his mother = $50 - 20 = 30 \text{ unit}$.

Electronics gadgets and monthly expenses from the remaining = $\left(\frac{3}{10} \times 30\right) = 9 \text{ unit}$

Remaining amount which transferred to his friend's account = 41000

⇒ (20 + 21) unit = 41000

⇒ 41 unit = 41000

⇒ 1 unit = 1000

⇒ 50 unit = 50000

∴ The monthly salary of Shiva is Rs. 50000.

∴ The correct answer is option (c).

29. (a) Given question,

Sachin * Madan – Reena

⇒ It means Sachin and Madan are of the same age and Reena is younger than Madan. The means that Reena is the youngest.

∴ The correct answer is option (a).

30. (b) As per the given question.

⇒ X + Y + Z means X is younger than Y and Y is younger than Z. This means that X is the youngest.

∴ The X + Y + Z is same as Z – Y – X

∴ The correct answer is option (b).

31. (c) Given series

56, 58, 62, 70, 84, 118, 182

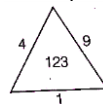
$\xrightarrow{+2}$ $\xrightarrow{+2^2}$ $\xrightarrow{+2^3}$ $\xrightarrow{+2^4}$

The pattern follows is $2 + 2^2 + 2^3, 2^4$

So, 84 is wrong and must be replaced by $(70 + 16)$ i.e. 86.

∴ The correct answer is option (c).

32. (c) As per the given question.



⇒ It follows the sequence.

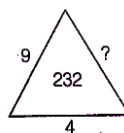
$1 \rightarrow 1^2$

$4 \rightarrow 2^2$

$9 \rightarrow 3^2$

So, in triangle it is written as 123.

In the same way.



⇒ It also follows the sequence

$4 \rightarrow 2^2$

$9 \rightarrow 3^2$

$4 \rightarrow 2^2$

So the missed term is 4.

∴ The correct answer is option (c).

33. (a) Given,

Students who were successfully appeared = 78%

Total number of failure = 176

Students who got first class = 34%

Students who failed = $(100-78)\% = 22\%$

22% of total students = 176

1% of total students = $\frac{176}{22} = 8$

34% of total students = $8 \times 34 = 272$

The number of students who got first class = 272

∴ The correct answer is option (a).

34. (d) Given question

1	7	9
2	14	?
3	105	117

The pattern follows in the series is:

⇒ $(2 \times 1 + 1) = (2 + 1) = 3$

⇒ $(7 \times 14 + 7) = (98 + 7) = 105$

⇒ In the same way

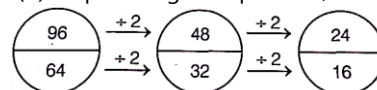
$(9 \times ? + 9) = 117$

$9x + 9 = 117$

$9x = 108 \Rightarrow x = 12$

∴ The correct answer is option (d).

35. (b) As per the given question,



∴ The correct answer is option (d).

36. (a) Cube has 6 sides.

As no two adjacent sides have the same colour, any pair of two opposites sides can have the same colour.

Hence, minimum number of colour required = $\frac{6}{2} = 3$.

∴ The correct answer is option (a).





37. (a) Let the correct time of train arrival be t and distance to reach the station be d .

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$1 \text{ min} = \frac{1}{60} \text{ h}$$

$$d = \frac{4 \times (t+6)}{60}$$

$$d = \frac{5 \times (t-6)}{60}$$

From Eqs. (i) and (ii)

$$\frac{4 \times (t+6)}{60} = \frac{5 \times (t-6)}{60}$$

$$5t - 4t = 30 + 24$$

$$t = 54 \text{ min}$$

Putting the value of ' t ' in Eq. (i)

$$d = \frac{4 \times (54+6)}{60}$$

$$d = 4 \times \frac{60}{60}$$

$$d = 4 \text{ km.}$$

$\therefore$ The distance covered by him to reach the station is 4 km.

$\therefore$ The correct answer in option (a).

38. (b) Let the initial value of fraction be x .

$$\text{Then new value of fraction} = \frac{5}{4}$$

$$\therefore \frac{5}{4} = x \times \frac{125}{0.8}$$

$$\therefore \frac{5}{4} = x \times \frac{25}{16}$$

$$z = \frac{5}{4} \times \frac{16}{25} \Rightarrow x = \frac{4}{5}$$

$\therefore$ The correct answer is option (b).

39. (d) Five friends Amar, Kapil, Sarvesh, Rohan and Nagesh.

Different colours Red, Yellow, Blue, White, Green

(White they were going in attend a party. These colours are not in order)

Different Hobbies: Reading, Playing, Outing, Singing, and Writing.

1. Kapil, who likes singing, does not wear Yellow shirt.
2. Sarvesh wears Red Shirt and he does not like reading or writing.
3. Nagesh likes playing and he does not wear Blue or Yellow shirt.
4. Amar likes Writing and Rohan does not wear Yellow or green shirt.

Friend	Colour	Hobbies
Amar		Writing
Kapil		Singing
Sarvesh	Red	
Rohan		
Nagesh		Playing

Amar wears Yellow shirt and Rohan likes Reading.

Sarvesh likes outing.

The colour of Kapil's shirt may be Green or Blue or White.

The colour of Rohan's shirt may be white or Blue.

The colour of Nagesh's shirt may be white or Green.

Rohan likes Reading

Friend	Colour	Hobbies
Amar	Yellow	Writing
Kapil	Green/Blue/White	Singing
Sarvesh	Red	Outing
Rohan	Blue/White	Reading
Nagesh	Green/White	Playing

The colour of Kapil's shirt can't be answer (d) as data is insufficient.

$\therefore$ The correct answer is option (d).

40. (b) The arrangement as per data given is

Friend	Colour	Hobbies
Amar	Yellow	Writing
Kapil	Green/Blue/White	Singing
Sarvesh	Red	Outing
Rohan	Blue/White	Reading
Nagesh	Green/White	Playing

$\therefore$ Amar likes writing.

$\therefore$ The correct answer is option (b).

